

Manohar Mishra

Senior Data Scientist

+91-6290040766

sirmisra@gmail.com

linkedin.com/in/sirmisra

PROFESSIONAL SUMMARY

Senior Data Scientist with ~6 years of experience spanning supply chain, logistics, advertising, and manufacturing domains. Deep expertise in Statistical Analysis, Predictive Modelling, Market-Mix Modelling, Mathematical Optimization, and Agentic AI systems. Proven track record of building production-grade ML solutions, autonomous multi-agent systems, and GenAI-powered applications using Snowflake, LLMs, and causal inference frameworks. Adept at translating complex analytical insights into measurable business outcomes.

CORE COMPETENCIES

Machine Learning & AI	Agentic AI Systems	Statistical Modelling	Operations Research	Market Mix Modelling
LLM Systems	Causal Inference	Data Analytics	Process Mining	Problem Framing

PROFESSIONAL EXPERIENCE

Kipi.ai / V4C.ai

Sep 2024 – Present

Data Scientist

Data & analytics firm specializing in Snowflake solutions and Generative AI consultancy (Dataiku).

- Built autonomous RCA agent (InsightOut) using Snowflake Cortex Code, reducing diagnostic time by >75% via causal inference and Monte Carlo simulation.
- Engineered a multi-agent data migration system to migrate data from legacy sources (SQL Server, Oracle) to Snowflake, with parallel execution, dynamic warehouse scaling, LLM-powered DDL conversion, PII detection, and automated reconciliation.
- Developed a PDF-to-Excel OCR tool in Snowpark, achieving >97% reduction in manual table extraction time.
- Engineered a Bayesian Market Mix Model, delivering 7.79% sales gain through optimized media budget allocation.
- Optimized Dataiku flows on Snowflake, reducing project costs by ~30%.
- Delivered multiple GenAI POCs using Python, Streamlit, and LangChain, showcasing RAG-based capabilities to enterprise clients.

Pitney Bowes Inc.

Apr 2022 – Sep 2024

Data Scientist

Global shipping and mailing company providing technology, logistics, and financial services.

- Automated sortkey-to-machine/bin mapping, reducing labor costs by 5–7% and eliminating manual planning days.
- Built ML classifier for restricted e-commerce export items, capturing 86% of restricted items using NLP on category descriptions.
- Enhanced Estimated Delivery Date model using CatBoost within Snowflake, improving forecast accuracy.
- Designed hypothesis testing engine with Plotly/Dash UI to analyze root causes of parcel delays.

- Developed process mining & anomaly detection system to trace parcel routes across warehouses using PM4PY and Cytoscape.
- Built container booking optimization model using Prophet to forecast volume and drive smart truck booking/cancellation decisions.

Tega Industries Ltd.

Dec 2020 – Apr 2022

Graduate Engineer Trainee

Leading manufacturer of wear-resistant lining components for mining and mineral processing industries.

- Performed regression analysis to identify key factors driving product wear rate, face angle, and height drop.
- Conducted statistical analysis of wear data for product lifecycle estimation, informing R&D and quality decisions.

QUANTIFIABLE IMPACT

- | | |
|---|--|
| • >75% reduction in diagnostic time via InsightOut autonomous RCA agent | • >97% time reduction in scanned PDF-to-Excel extraction |
| • 7.79% sales gain through Market Mix Model budget optimization | • ~30% reduction in project cost via Snowflake & Dataiku optimization |
| • 5–7% labor cost reduction via efficient sortplan algorithm | • 86% capture rate of restricted items using ML classifier |

AGENTIC AI PROJECTS (SELF-INITIATED)

InsightOut: Autonomous RCA & Prescriptive Analytics Agent

Technologies: *Snowflake Cortex Code (CoCo), Snowpark (Python), SQL, Causal Inference (OLS/Shapley), Monte Carlo Simulation, Optimization Models*

Built an autonomous agent that identifies statistically significant root causes of business anomalies using causal inference (OLS regression & Shapley values). Recommends optimized, risk-quantified corrective actions through large-scale Monte Carlo scenario simulation, reducing diagnostic time by >75% compared to manual RCA processes.

Autonomous Data Migration Agent (Snowflake Cortex)

Technologies: *Snowflake Cortex Agents, Snowpark Python, Multi-Agent Architecture, SQL, YAML, ThreadPoolExecutor*

Designed and built an intelligent multi-agent system to automate end-to-end data migration from heterogeneous legacy sources (SQL Server, Oracle) to Snowflake. Features include a Master-Worker parallel execution architecture using ThreadPoolExecutor, dynamic warehouse scaling based on table row count, LLM-powered DDL conversion, automated PII detection and masking, and self-validating data reconciliation post-migration.

KEY PROJECTS

Estimated Delivery Date

Technologies: *Python, Snowflake, CatBoost*

Built and enhanced an Estimated Delivery Date model for a global logistics network, using CatBoost regression within Snowflake to improve prediction accuracy for parcel arrival times across the shipping network.

SortPlan Optimization

Technologies: *Python, PYOMO, Mixed Integer Programming*

Automated the mapping of sortkeys to machines and bins using Mixed Integer Programming, eliminating manual planning overhead and reducing labor cost by 5-7%.

Identify Export Control Classification Number

Technologies: *Python, Snowflake, scikit-learn, LinearSVC, NLP*

Built an NLP-based ML classifier to identify restricted and non-restricted cross-border e-commerce items using category descriptions, capturing 86% of restricted items and enabling faster, compliant shipment processing.

Cause-Effect Analysis

Technologies: *Python, SciPy, Plotly, Dash, Hypothesis Testing*

Designed a hypothesis testing engine to assess the impact of automation exceptions on parcel delays, with an interactive Dash UI for operations teams to explore root causes.

Process Mining & Anomaly Detection

Technologies: *Python, PM4PY, Dash, Cytoscape, Jinja2*

In-depth analysis of parcel event data across warehouses and machines to identify dominant routes and investigate abnormal paths, with an interactive network visualization built using Dash and Cytoscape.

Market Mix Modelling

Technologies: *Python, PyMC, Snowpark*

Bayesian Market Mix Model to measure effectiveness of media channels and campaigns, with optimal budget allocation recommendations that delivered a 7.79% gain in sales.

Container Booking Optimization

Technologies: *Python, Prophet*

Developed a container volume forecast using Prophet and built a rule-based decision engine to guide truck booking and cancellation decisions in advance, reducing net logistics cost.

SKILLS

Languages	Python, SQL, NoSQL
ML Algorithms	Logistic Regression, SVM, Decision Tree, Gradient Boosted Trees, Random Forest, CatBoost, XGBoost, Ridge & Elastic-Net Regression, Prophet, KNN
Agentic AI	Snowflake Cortex Agents, Multi-Agent Architecture, LangChain, Llama-Index, Causal Inference, Monte Carlo Simulation
Frameworks	TensorFlow, PyTorch, PyMC, PYOMO, PM4PY, Streamlit, Plotly/Dash
Platforms	Snowflake, Snowpark, Dataiku

EDUCATION

Jadavpur University2016 – 2020

CGPA: 7.8 / 10

CERTIFICATIONS

- Applied Data Science Lab – WorldQuant University: Eight end-to-end data science projects covering SQL/NoSQL access, ML modelling, ETL pipelines, and data visualization.
- Inferential Statistics – Coursera: Hypothesis testing, p-value interpretation, and statistical analysis with Python.
- Data Visualization with Python – IBM: Interactive dashboards, chart selection, and Python visualization libraries.
- Master SQL for Data Science – Udemy: Complex queries, advanced SQL techniques, and 100+ real-world puzzles.
- COF-C02 SnowPro Core Certification – Snowflake: Validates core competency in Snowflake platform architecture, data loading, performance optimization, and security.